## 12.4.1 Getting an item by key

Let's start with a simple example. You want to find out the contact details about Emma, which are stored in the `todo-user` table.

The simplest form of data retrieval is looking up a single item by its primary key, for example, a user by its ID. The `getItem` SDK operation to get a single item from DynamoDB can be used like this.

**Listing 12.7 DynamoDB: Query a single item (index.js)**

```
const params = {
  Key: {
    attr1: {S: 'val1'}                    ①
  },
  TableName: 'app-entity'
};
db.getItem(params, (err, data) => {       ②
  if (err) {
    console.error('error', err);
  } else {
    if (data.Item) {                      ③
      console.log('item', data.Item);
    } else {
      console.error('no item found');
    }
  }
});
```

① Specifies the attributes of the primary key

② Invokes the getItem operation on DynamoDB

③ Checks whether an item was found

The `user <uid>` command retrieves a user by the user's ID. The code to fetch a user is shown in listing 12.8.

**Listing 12.8 nodetodo: Retrieving a user (index.js)**

```
const mapUserItem = (item) => {                    ①
  return {
```